Simulation Milestone 01
Autonomous Systems Project

Team 19

I. Introduction

This milestone demonstrates the successful setup of the simulation environment using ROS2 Jazzy and Gazebo.

II. Simulation Environment Setup

The simulation environment was implemented using ROS2 Jazzy running on Ubuntu 24.04. A ROS workspace was created and the Gazebo simulator was integrated with ROS using the `ros_gz` packages.

III. Vehicle Topics and Communication

Several topics were discovered during the inspection of the simulation environment. The most important topics related to the vehicle are summarized in Table I.

TABLE I
Important Gazebo Topics for the Vehicle

Topic	Message Type	Description
/model/vehicle_blue/cmd_vel	gz.msgs.Twist	Velocity command
/model/vehicle_blue/odometry	gz.msgs.Odometry	Position,velocity feedback

The topic `/model/vehicle_blue/cmd_vel` is used to send motion/velocity commands to the vehicle. This topic uses the `gz.msgs.Twist` message which contains both linear and angular velocity components. This topic is used for controlling the car, If you publish to it, the car will move with the linear and angular velocities you provided.

The topic `/model/vehicle_blue/odometry` provides feedback about the vehicle's state, including its position, orientation, linear velocity and angular velocity within the simulation world. It publishes the vehicle's state.

IV. Vehicle Motion Validation

To verify that the simulation environment is functioning correctly, a velocity command was published to the vehicle command topic:

```
gz topic -t /model/vehicle_blue/cmd_vel \
-m gz.msgs.Twist \
-p "linear: {x: 1.0} angular: {z: 0.2}"
```

After executing this command, the vehicle started moving in the Gazebo simulation environment. The odometry topic was monitored using the following command:

```
gz topic -e -t /model/vehicle_blue/odometry
```

The resulting messages showed continuous updates of the vehicle position and velocity, confirming that the vehicle responded correctly to the command input.

V. Vehicle Model in Gazebo

Figs. 1 and 2 show the vehicle model named `vehicle_blue`, which represents a car-like robot in the Gazebo simulation environment.

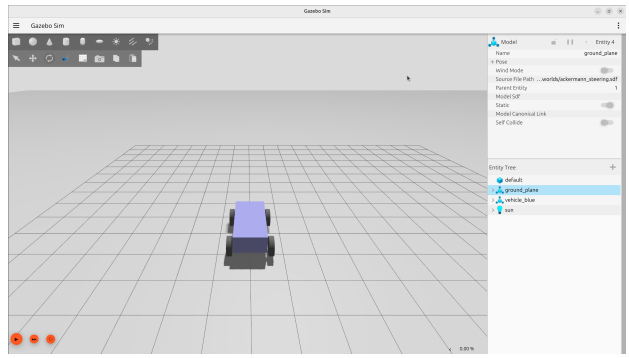

Fig. 1. Vehicle Model in Gazebo

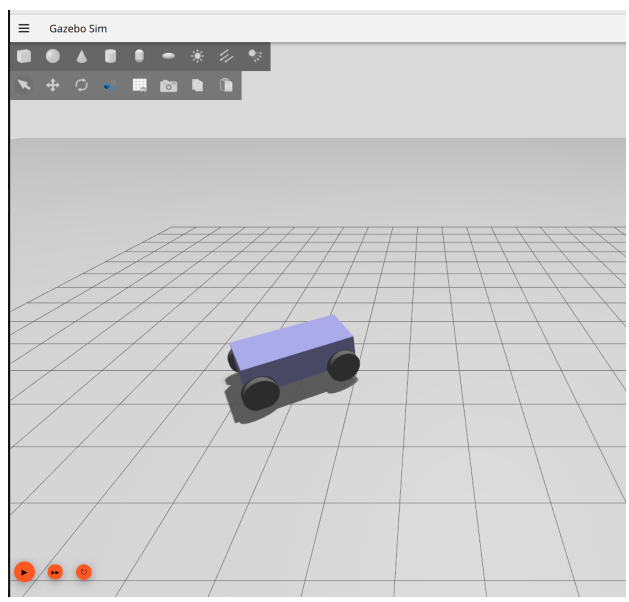

Fig. 2. Vehicle Model in Gazebo